

K-101 Centrifugal Compressor Maintenance & Vibration Report

Confidential Industrial Diagnostics • Mangalore Refinery Hydrocracker Complex

1. Equipment Identification & Operating Conditions

Equipment Tag: K-101 Centrifugal Gas Compressor. Location: Hydrocracking Unit 2.
Service Fluid: Light Hydrocarbon Mixture. Design Suction Pressure: 24.5 bar, Discharge Pressure: 72.0 bar.
Normal Operating Speed: 10,450 RPM. Primary Drive: High-pressure condensing steam turbine.

2. Observed Vibration Excursions & Telemetry Analysis

During routine seismic accelerometer survey on Drive End (DE) bearing housing, overall radial vibration was recorded at 6.4 mm/s RMS (Velocity).
This reading represents a 70% increase above the baseline benchmark (3.2 mm/s RMS) and exceeds the ISO 10816-3 Zone C (Unrestricted Operation Limit) threshold of 4.5 mm/s RMS. Spectrum FFT analysis shows dominant 1X running frequency component with prominent 2X harmonics, indicative of angular shaft misalignment and early bearing race spalling.
Non-drive end (NDE) bearing recorded moderate vibration at 3.9 mm/s RMS with elevated lube oil return temperature (68°C vs 55°C standard).

3. Mechanical Seal & Lube Oil Inspection

Dry gas seal leakage rate increased from 1.2 Nm³/hr to 3.8 Nm³/hr on the primary vent line, indicating secondary O-ring elastomer degradation.
Lube oil spectrochemical analysis revealed elevated tin (Sn: 18 ppm) and copper (Cu: 12 ppm) particulate concentrations, confirming babbitt bearing wear.
Oil particle count: ISO 4406 cleanliness rating degraded to 21/18/15.

4. Required Corrective Actions & Preventive Maintenance

IMMEDIATE ACTION: Reduce compressor load by 15% to maintain vibration below 5.5 mm/s until shutdown.
SCHEDULED OUTAGE: Execute laser shaft realignment between turbine driver and compressor coupling.
REPLACEMENT: Disassemble DE journal bearing, inspect tilting pads for babbitt wipe, and install refurbished dry gas seal cartridge (Part #DGS-K101-REV4).
HUMAN VERIFICATION: Final alignment tolerances must be signed off by Lead Rotating Equipment Engineer.